

9 June 2026

Dear Editors,

We are pleased to submit this revision of our paper on the use of cross-lagged panel models in life course and development research and criminology more generally. We were pleased with the positive reception and engaging feedback. Below we provide point-by-point responses for each paragraph of Reviewer 1's thoughtful comments.

The paper overall provides a nice discussion of key issues when considering applying CLPMs to understand reciprocal relationships in criminological research. The paper focuses on two 'case studies' – questions relating to employment and fear of crime, which helps ground the technical side of the paper in real empirical research. There are some areas that this could be strengthened so that the paper does not get too technical for readers. My comments highlight some of these places where this can be done, as well as some additional comments to consider regarding the role of time.

Yes, we appreciate the suggestion to avoid getting overly technical; we have a tendency to let our excitement over technical details spill out.

First, the two use cases are focused on different kinds of reciprocal relationships, which I agree is an important issue. I do think that the theoretical and empirical underpinnings of these reciprocal relationships could be elaborated on more for the reader. For example, what does existing research show regarding the effect of criminal behavior on job loss? Does not accounting for this affect the estimates in empirical work? In relation to disorder, what is the evidence for each direction and how have studies examined this reciprocal relationship in previous work (including CLPMs but perhaps also beyond). There is quite a growing area of research on momentary measures of fear of crime (e.g., Solymosi, Kronkvist, Chataway, etc.) that may help further elaborate on the temporal, reciprocal processes. Incorporating existing empirical work for both use cases can help the reader contextualise and situate the problem in ongoing work, and help understand how these main issues can affect our interpretation of the evidence.

We appreciate the suggestion to get deeper into the theoretical processes and relate empirical evidence. Our goal is to use these examples to illustrate methods and motivate readers to be concerned with reciprocity broadly rather than to address methodological problems or outstanding debates in these substantive areas. As a result, we have expanded the introduction of the substantive issues but still largely deferred to the literature for readers interested in the substance (pp. 5-6).

The authors discuss the reciprocal relationship between employment and crime, but I think this requires a bit more elaboration and perhaps some consideration of the underlying theoretical process. First, they state that the reciprocal relationship (crime->employment) is not usually of substantive interest (pg 9), but in my reading I am not sure that is a correct assumption. In the desistance process, which can usually be messy and intermittent, the influence of crime on employment reflects a substantive part of the process. Engagement in crime, or even intentions or openness to crime can disrupt the embeddedness and building of social capital, controls, and commitment to conventional society/workplace. The authors should perhaps reflect on whether (theoretically) this is merely something to be accounted for in the model. More connection with desistance research can perhaps help here.

We agree that it is inaccurate to state that theoretical reciprocity is not of substantive interest to most criminologists and have corrected this. We have also edited this section (p. 6) to emphasize that, like

estimands, viewing any given reciprocity as reverse causality or theoretical reciprocity is a choice linked to theory and research questions.

In addition, I wondered if the temporal processes connecting employment->crime and crime->employment are equivalent such that one could examine them in a CLPM. A CLPM fundamentally assumes that the temporal order, timing, and process of change is the same for both directional processes (or at least can be captured in the same modelling strategy and time interval). While the employment->crime process is described as gradual and slow-moving, I'm not sure the crime->employment process can be described in the same way, as job loss is a discrete event (although the measure stated is proportion of time employed). This could be clarified by being a bit more explicit about the underlying processes and assumptions, for example by connecting to theoretical and empirical work (as mentioned in the comment above).

The substantive examples here are meant to serve as illustrations for the methodological considerations rather than as a focus of the paper. As a result, we have instead broadened the discussion of crime -> employment effects to include substitution, human capital, and job acquisition to avoid dwelling on job loss and to include processes operating at different paces (pp. 5-6).

The authors mention continuous time models, and I think more attention could be paid to assumptions and specifications of discrete vs continuous time, and the implications for modelling and inference. Many readers may not be familiar with continuous time, or that we make implicit assumptions about discrete vs continuous time when using panel models in these cases.

We thank the reviewer for prompting us to develop this point. We have now included an expanded discussion of continuous-time models (p. 11) that (i) makes explicit the normally implicit temporal-scale assumption of discrete-time CLPMs, (ii) introduces the continuous-time alternative and its main advantage (interval-comparable parameters), and (iii) notes the embedding problem as a limit on what continuous-time models can recover from sparse panel data. We agree this is an important consideration that many applied readers will not encounter elsewhere, but we have kept the treatment compact so as not to redirect the paper toward continuous-time methods specifically.

The authors discuss measurement a few times throughout, distinguishing between retrospective and current measures. This is an important point, and I think the authors could elaborate even more on these issues. Retrospective measures often have different time frames, and "current" measures are sometimes framed in a general way that it could cover a much broader timeframe than what people feel in the moment. In particular for CLPMs, the retrospective measures in different timeframes could create further issues in temporal ordering (eg, last 7 days, last 6 months). Retrospective measures also create multiple potential causal pathways within the CLPM (see Speyer et al: <https://www.tandfonline.com/doi/full/10.1080/00273171.2024.2428222>).

We have added text to the section introducing instantaneous and period measures (pp. 13-14) that (i) acknowledges retrospective windows can vary across measures within a single wave, (ii) notes that "current" perception items are themselves often interpreted by respondents over an unspecified window, and (iii) flags that mismatched windows compound the within-wave ambiguity we discuss elsewhere.

In the second half of the paper, the authors make several important points. However, sometimes the language is abstracted and technical referring to causal and statistical language, making it difficult to follow for a reader not very familiar with these issues. It depends on who the audience is for this paper, but I would guess that the authors want to aim at empirical researchers who may be interested – either theoretically or empirically – in examining reciprocal relationships. This audience may or may not have a strong statistical background. In the front end, the authors used the two theoretical case studies, but they are not applied as much in the second half of the paper. I would recommend that the authors bring their examples to the second half of the paper to illustrate the problems and solutions, and try to avoid using only technical language. Applying the issues/solutions to the case studies can help illustrate exactly how these would look in theory and empirical research.

This is a perhaps unfortunate consequence of having worked backwards, beginning with the technical issues we wanted to discuss and then building a scaffold around them. In response, we have made a sweep of the

paper looking to simplify and smooth out the technical discussion for readers by condensing some sections and moving some technical material to footnotes (e.g., p. 17-18) and integrating examples using our substantive topics as noted below.

Related to the above comment, the authors introduce two extensions to CLPM. It would be helpful for the reader if the authors provide more examples of criminological studies who have applied these methods. If possible, it would also be helpful to discuss how these models/findings compare to traditional CLPM findings – although that may be a whole new study in itself, so feel free to disregard. At least some citations of examples can help the reader see these models applied to questions in criminology.

We agree that seeing some papers applying these methods is a great addition; we have added a number of examples of studies applying these methods in the development and life course subfield and in other areas (e.g., p. 18 and p. 20), admittedly including some of our own work which explicitly demonstrates the suggestions made in the present work.

On page 19, again I feel it is a bit too abstracted to get a sense of how this problem looks in empirical literature. It may be that it is difficult to recognize, but as a reader I am wondering if and when issues with temporal order have led to mixed or unexpected results in criminological research. Can the authors provide some empirical examples of this problem?

On page 20, I have the same comment. The authors at one point state “These issues are common in developmental studies of individuals...etc” and provide a citation, but I was left wondering about those common issues, what do they look like? Can the authors provide some concrete examples in the text alongside the discussion of the more technical issues?

We have expanded a worked illustration of the Vaisey–Miles sign-flip result (p. 21), drawing on Ousey, Wilcox, and Fisher's (2011) study of the victimization–offending relationship. We chose this example because it provides clear evidence of how the misspecification problem manifests in published criminological work. We frame the example to emphasize that the issue reflects the limits of methodological knowledge at the time of the original study rather than researcher error.

For the inter-temporal variation section, we have added a life-course criminology example (p. 23). Working within our employment running example, we note that the foundational within-person evidence for turning-point effects required decades of data (Sampson, Laub, and Wimer 2006), and we contrast this with register-based studies whose finer resolution reveals these processes to be gradual and even anticipatory (Skardhamar and Savolainen 2014; Skardhamar, Monsbakken, and Lyngstad 2014). We use this to show that short panels can contain too little within-person change to identify a slowly-unfolding effect.

Overall, bringing in more examples from theory and empirical work can help ground these issues and (in my view) reach a broader audience. The first half of the paper does a good job at connecting these issues with theory and research, and so the authors should simply continue that throughout the paper.

We agree completely with this general thrust of these comments and agree that grounding the back end of the paper will make it connect better with our intended audience.

We hope you find these revisions to be a suitable response to the recommendations above. Thank you again for the thoughtful comments.

Yours sincerely,

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